Lecture 04: Extended Entity-Relationship Diagram (EERD)

ITCS216 Fundamentals of Database Systems

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This lecture is adopted from ITCS241 content in 1/2022 and collarborate with Asst. Prof. Dr. Preecha Tangworakitthaworn

Learning Objectives

- After completing this lesson, you will be able to:
  - Describe the extended entity relationship (EER) model constructs and how they are represented in ERDs and EERDs.
  - Use entity clusters to represent multiple entities and relationships in the Extended Entity Relationship Diagram (EERD).

Why EER?

- **ER** model is useful to describe “traditional” database system.
- Complex data objects are difficult to model due to advanced requirements, for example:
  - Hierarchical structure of complex data object, or
  - Modern data object, such as, sound or multimedia data, or
  - Advanced data technology, such as, Geographic Information System (GIS).
- Semantic data modeling needs to be incorporated !
  - Result solutions → Extended Entity Relationship (EER) model.

Aviation Business

Common characteristics

Employee

- Firstname
- Lastname
- HiredDate



Pilot | Mechanic | Secretary | Accountant | DBA

- License
- Rating
- MedType

Only Pilot participate in some relationships that are unique to their qualifications. E.g. “employee flies airplane”

Unique characteristics

Let's see some PROBLEM…

Lots of "NULL" values (called Sparse Data)

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_LICENSE	EMP_RATINGS	EMP_MED_TYPE	EMP_HIRE_DATE
100	Kolmycz	Xavier	T				15-Mar-88
101	Lewis	Marcos		ATP	SEL/MEL/Instr/CFII	1	25-Apr-89
102	Vandam	Jean					20-Dec-93
103	Jones	Victoria	R				28-Aug-03
104	Lange	Edith		ATP	SEL/MEL/Instr	1	20-Oct-97
105	Williams	Gabriel	U	COM	SEL/MEL/Instr/CFI	2	08-Nov-97
106	Duzak	Mario		COM	SEL/MEL/Instr	2	05-Jan-04
107	Diante	Venite	L				02-Jul-97
108	Wiesenbach	Joni					18-Nov-95
109	Travis	Brett	T	COM	SEL/MEL/SES/Instr/CFII	1	14-Apr-01
110	Genkazi	Stan					01-Dec-03

Mahidol University
Wisdom of the Land

So how to solve this problem?

Hierarchical Structure (Specialization/Generalization)

Hierarchical Structure

- Used to demonstrate the concept of **super-types** and **sub-types.**
- Relationships described in terms of **IS-A or Type-of** relationships.
- Every subtype has one supertype to which it is directly related.
- Subtype exists only within context of supertype.
- Supertype can have many subtypes.

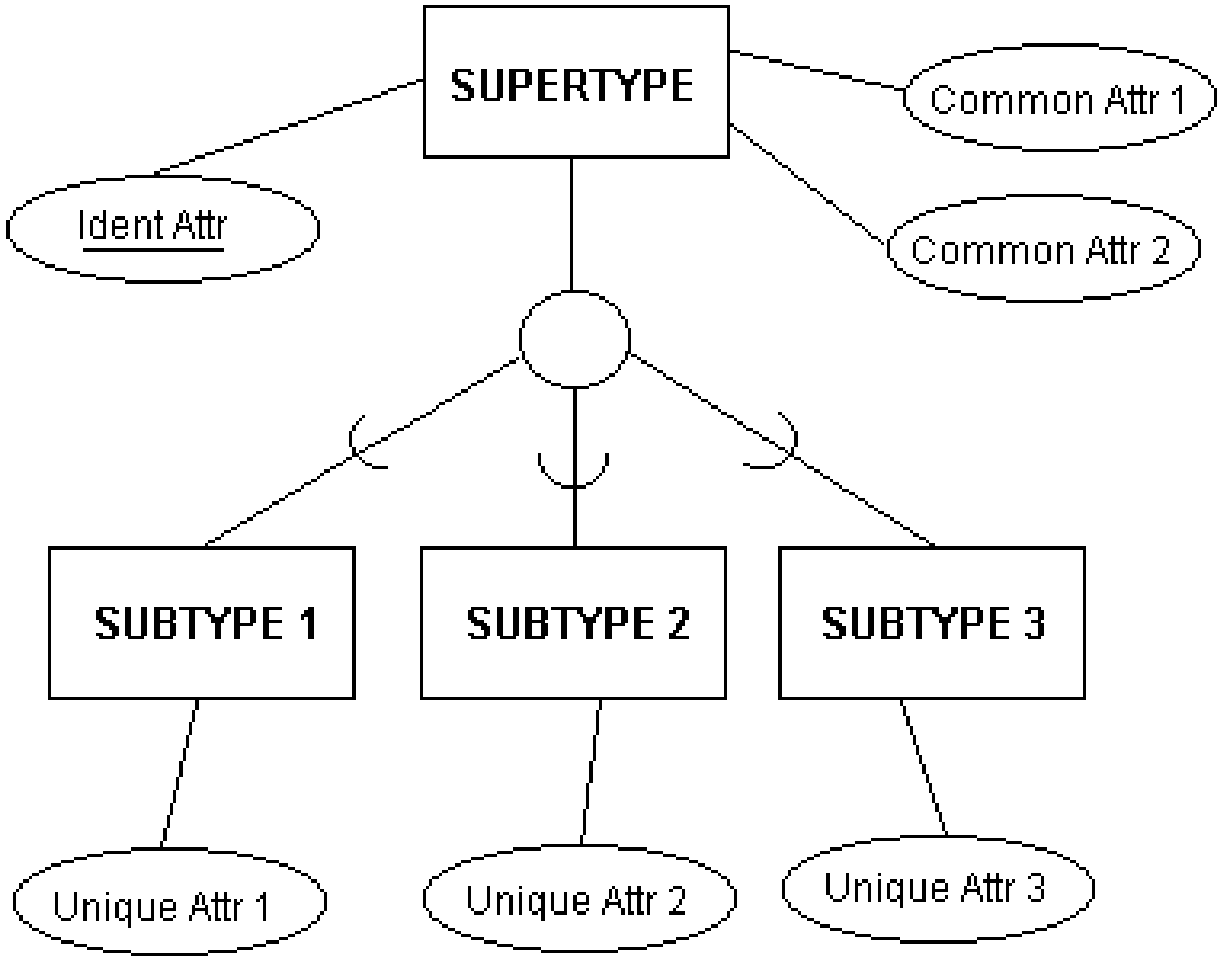
Entity Cluster

- Entity cluster is a set of one or more entity types and associated relationships grouped into a single hierarchical structure.
- Entity cluster formulates the hierarchical structure of EERD.

Entity Supertype and Subtype

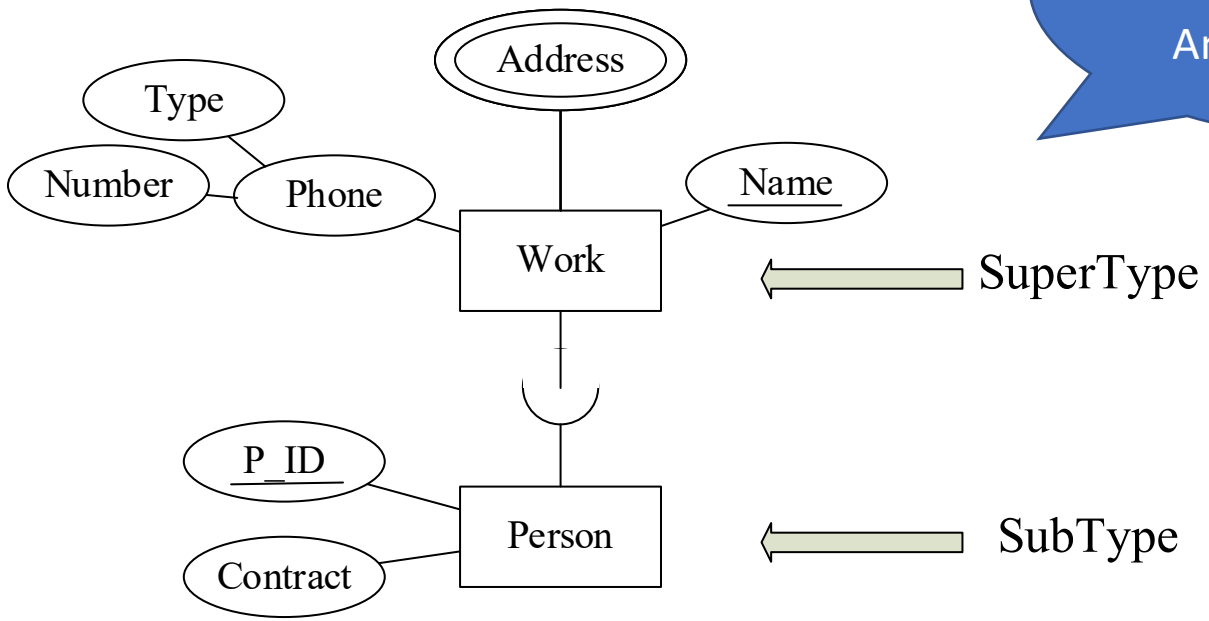
- **Entity Supertype**
  - Generic entity type related to one or more entity subtypes.
  - Contains common characteristics.
- **Entity Subtype**
  - Contains unique characteristics of each entity subtype.
- Example
  - Mammal – Human and Chimpanzee
  - Bank account – Saving and Current

Hierarchical Structure



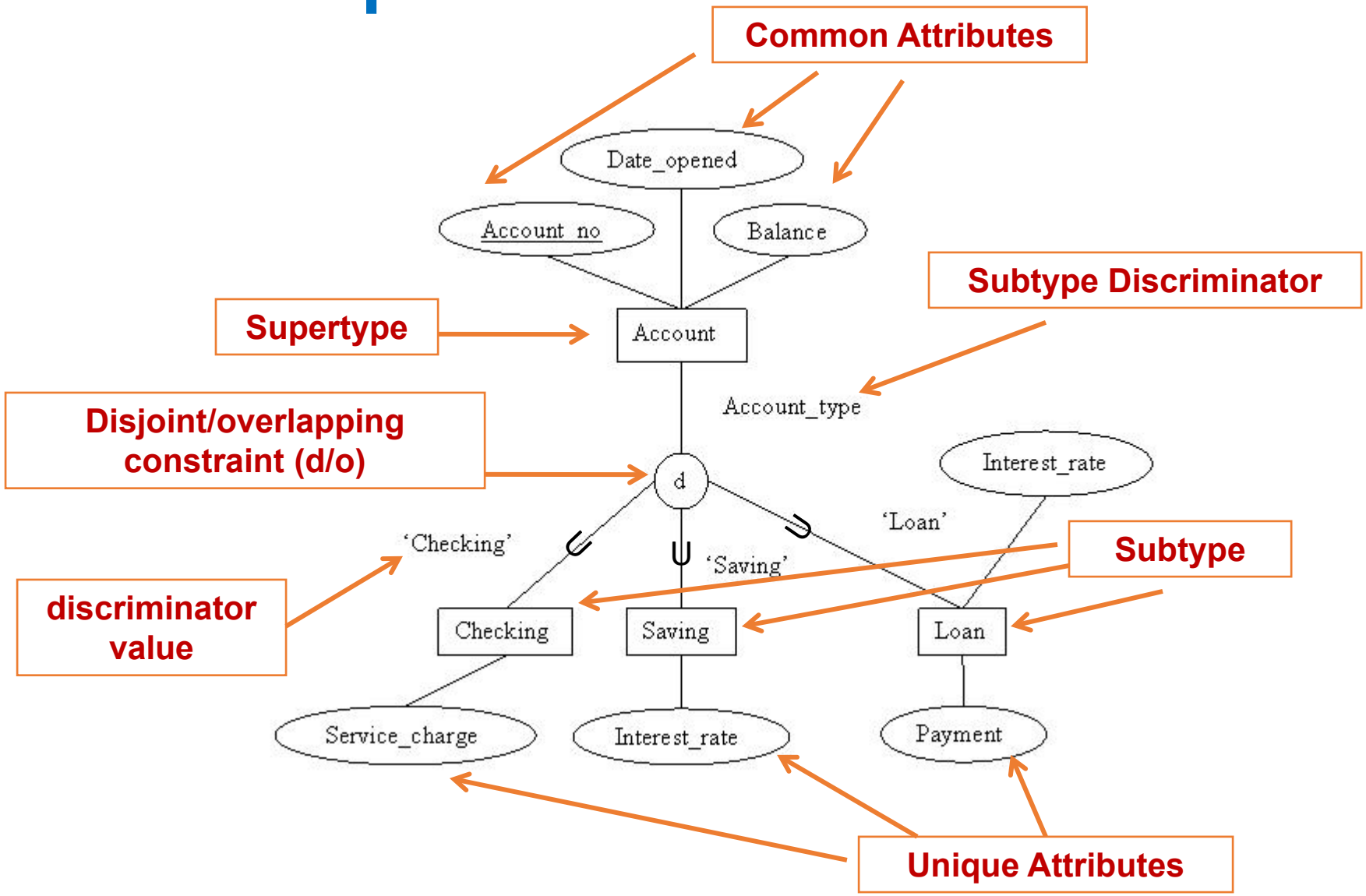
Example of EER

How many entity cluster?
Answer: only 1 cluster

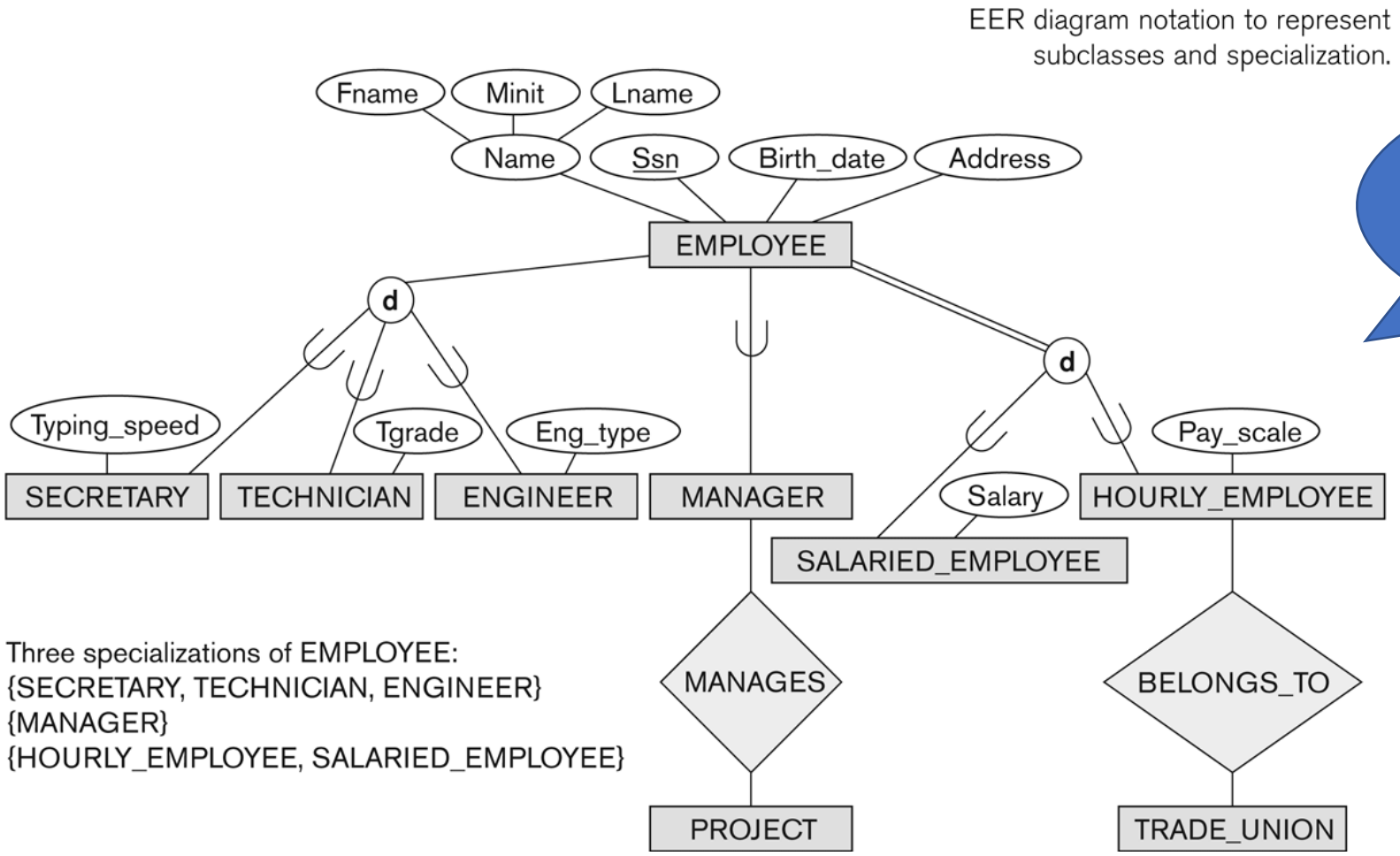


We clarify this hierarchical structure from determining the Arrow Direction.

EER Components



Example of EER



How many entity cluster?
Answer: 3 clusters

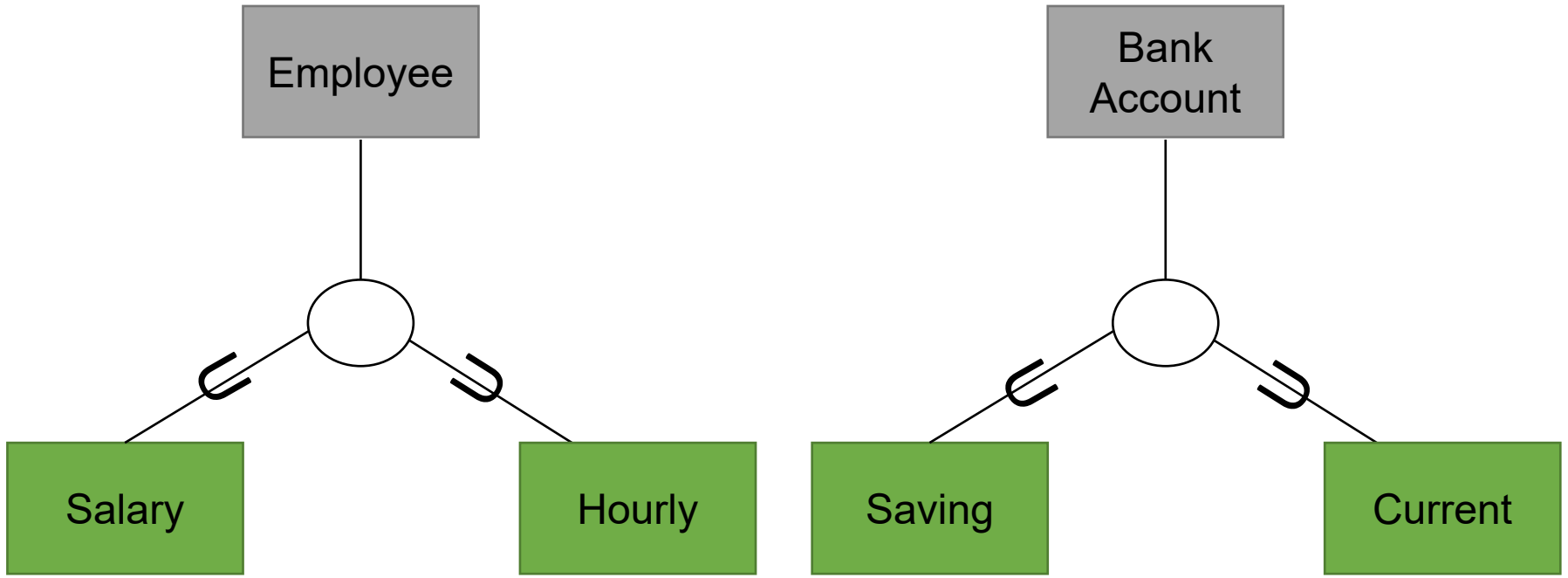
How to design EER

1. Identify Supertype entities and Subtype entities.
2. Identify common attributes.
3. Identify unique attributes.
4. Specify subtype discriminator and discriminator values.
5. Identify discriminator constraints.

1. Identify Supertype entities and Subtype entities

- This is the concept of categorizing or generalizing.
- Determine the terms: “Type of”, “Kind of”, or “Sub-type of”, e.g., Saving is a type of Bank Account.
- **Example**
  - **Bank Account** → Saving & Current
  - **Employee** → Salary & Hourly
  - **Student** → Postgraduate & Undergraduate

1. Identify Supertype entities and Subtype entities– (2)



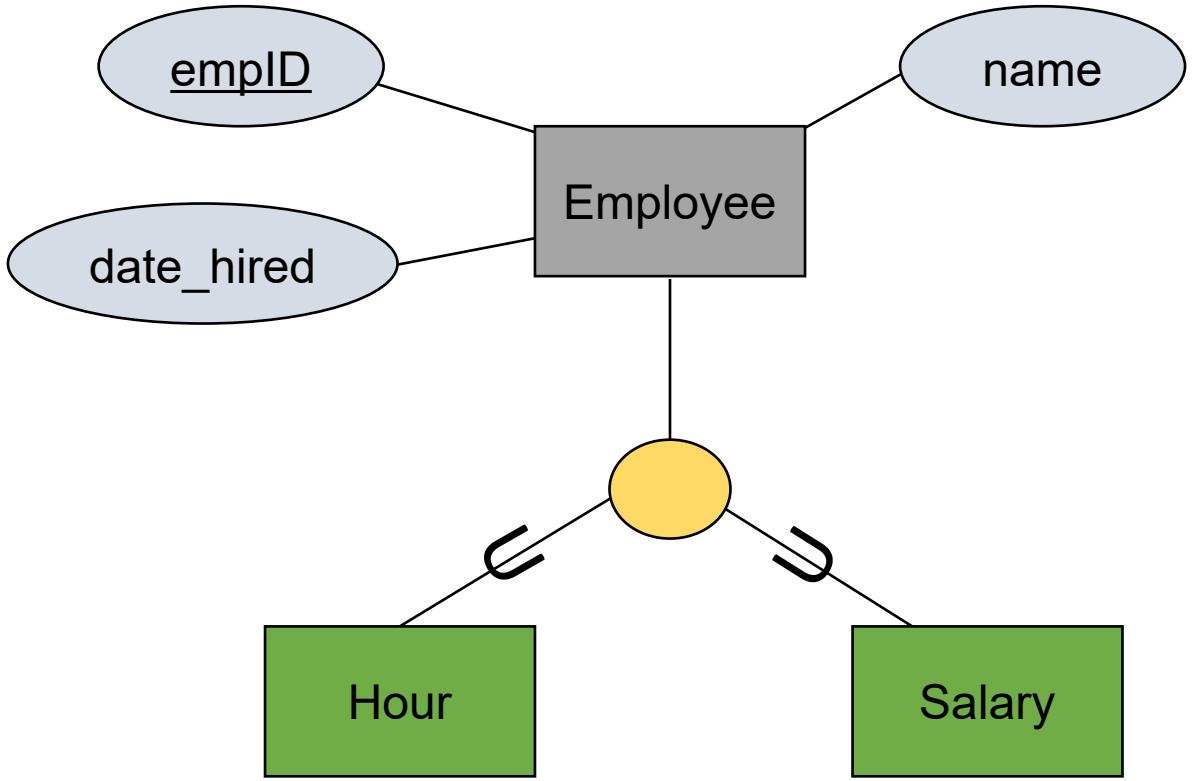
2. Identify common attributes

- **Common attributes** – shared attributes that belongs to both supertype and subtype.

- **Example**
  - For both saving and current accounts, they both have *name*, *surname*, *address*, *telephone*, *balance* of customers.

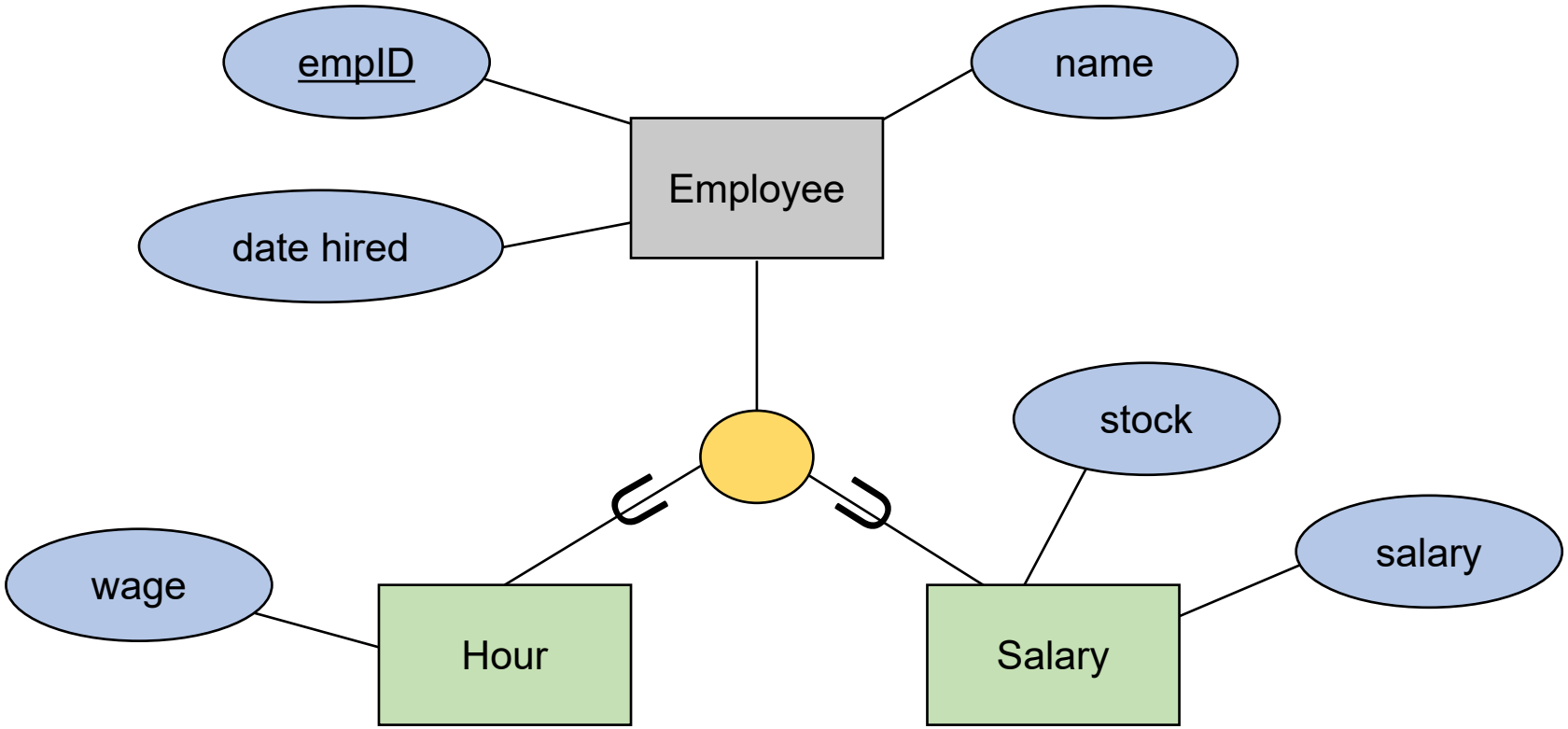
  - For salary and hourly employee, they both have *name*, *surname*, *address*, *telephone*, *date hired* of employees.

2. Identify common attributes – (2)



3. Identify Unique Attributes

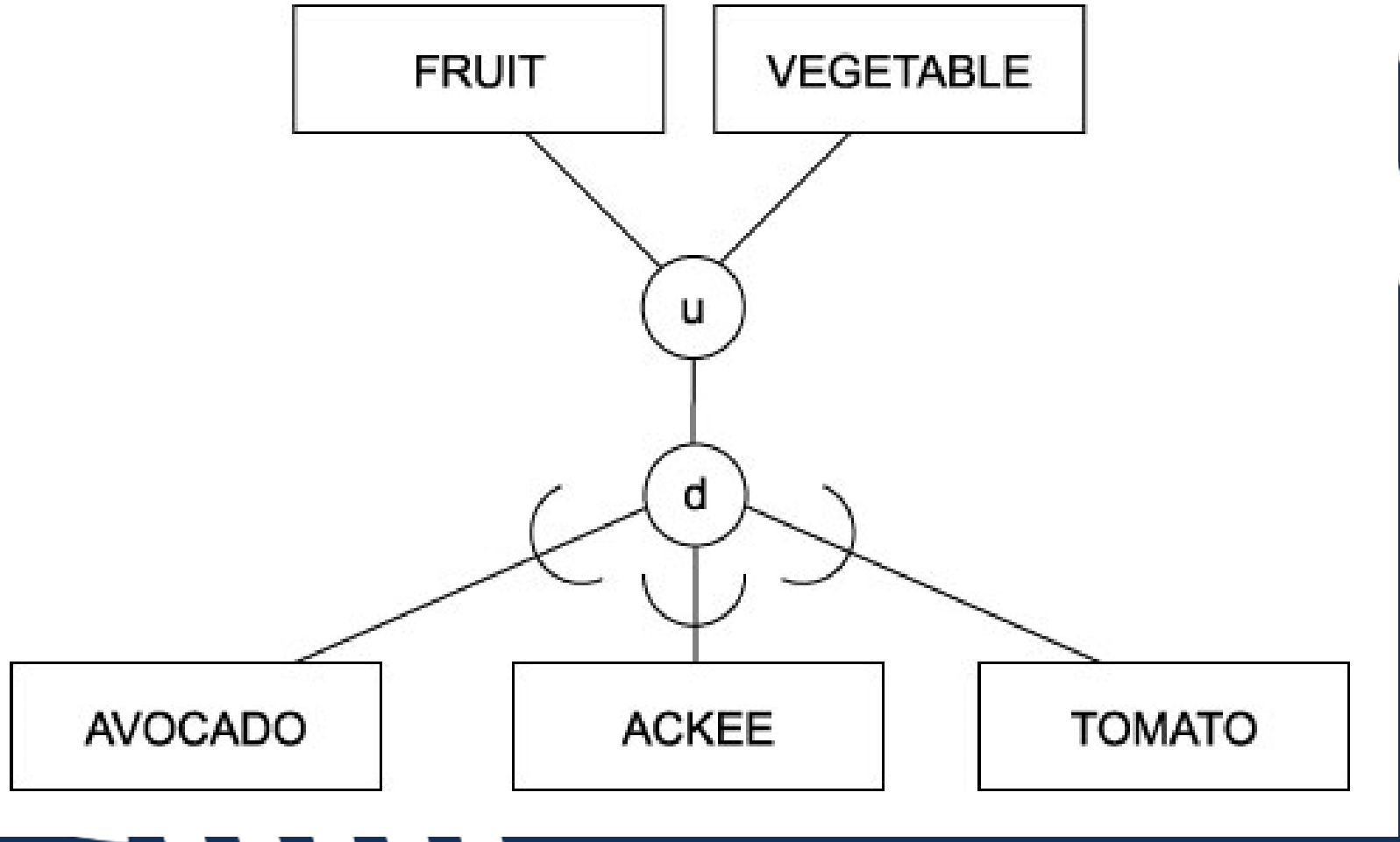
- **Unique attributes** – attribute(s) that belong to each particular subtypes, not share with the others. (Noted that all subtype entities should have at least one unique attribute.)



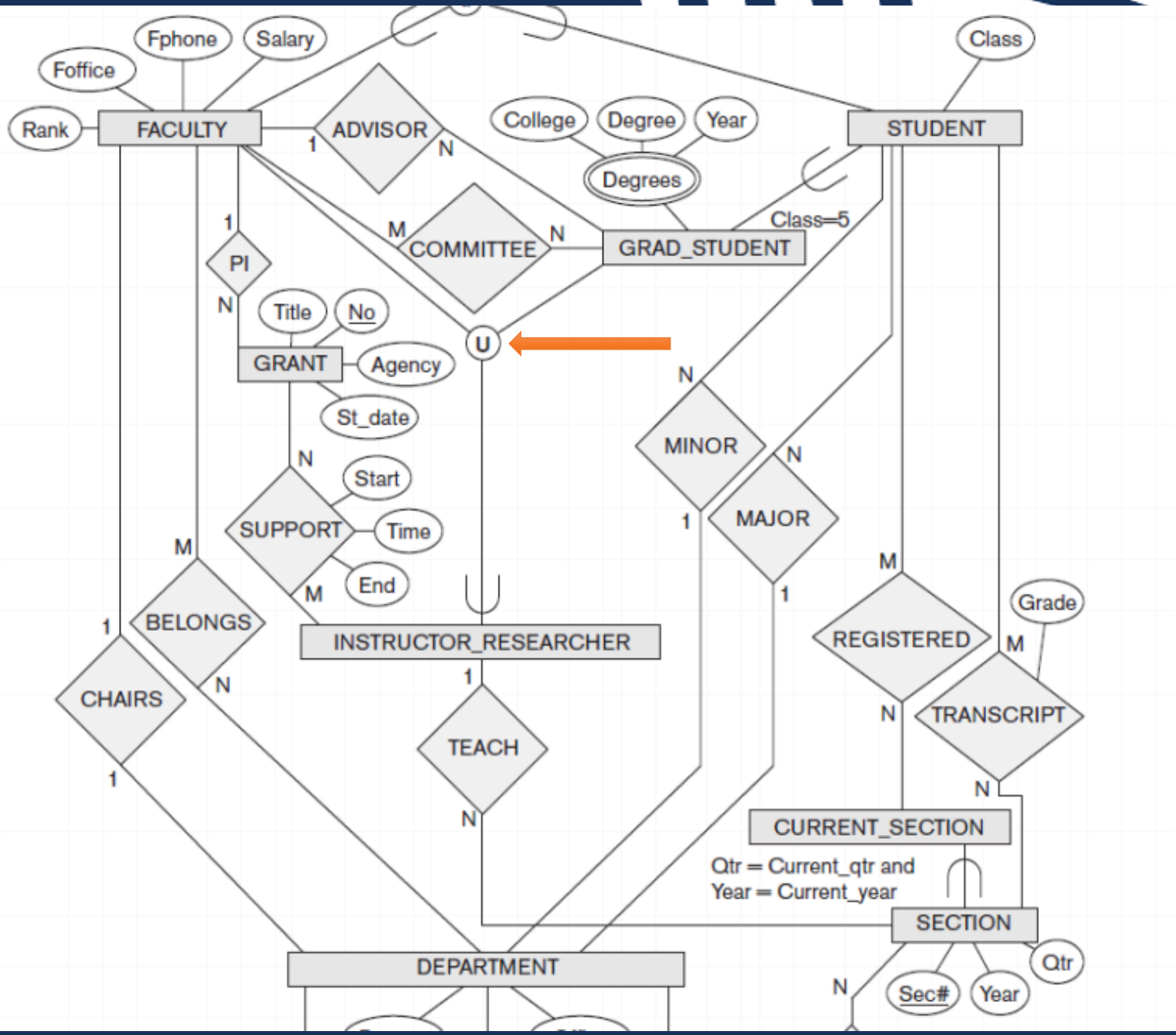
Inheritance

- Subtype Entities will inherit the following from Supertypes:
  - Primary key,
  - Attributes,
  - All relationships in which all upper level supertype entity participates.
- At implementation level, supertype and its subtype(s) maintain a 1:1 relationship.

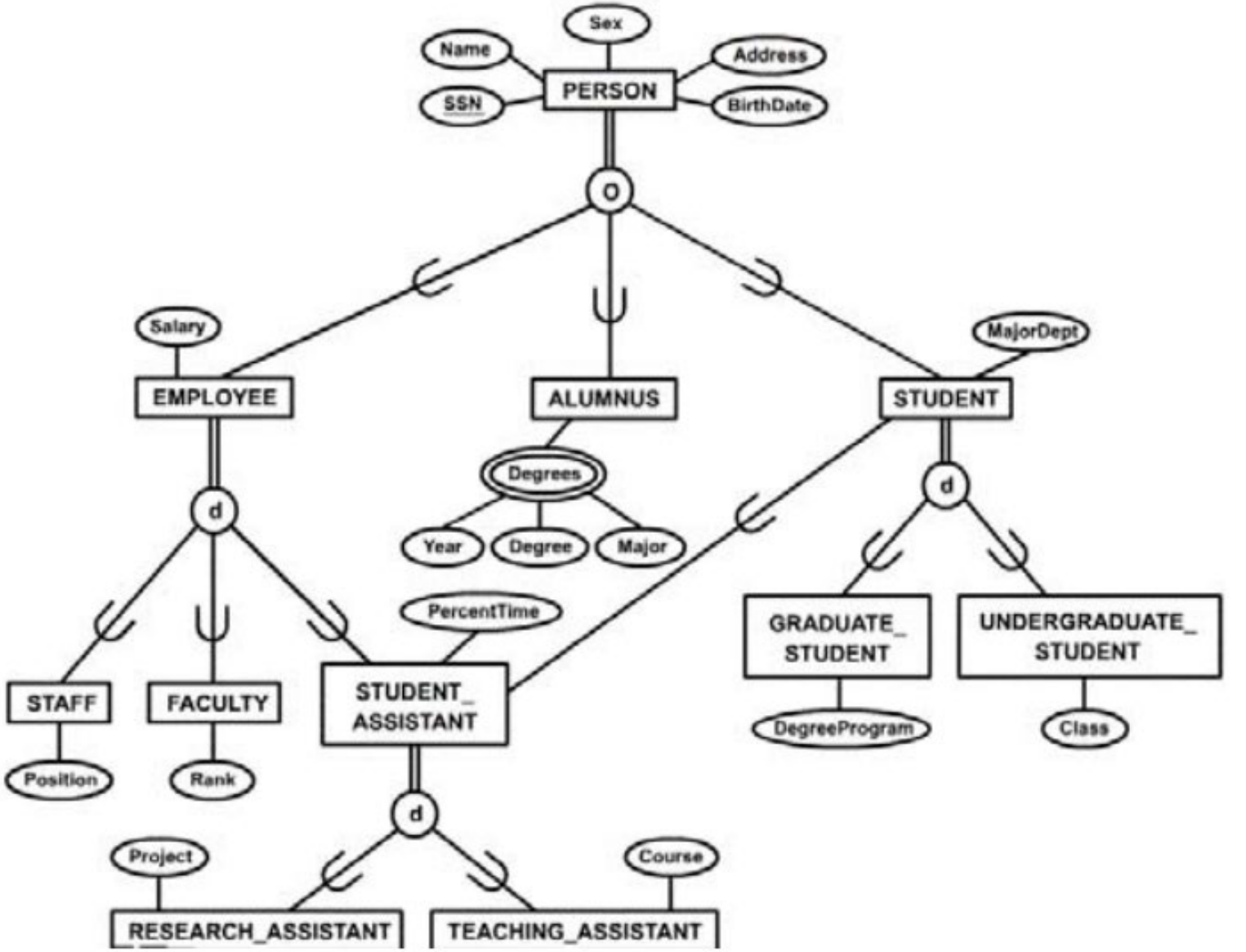
Multiple Inheritance



u: Union – a subclass/super-class relationship can have more than one super-class, and the subclass inherits from at most one of the super-classes (i.e. the subclass INSTRUCTOR_RESEARCHER will inherit the relationships and attributes associated with either FACULTY or Grad_STUDENT, but not both). Each super class may have different primary keys, or the same primary key. All members of the super-classes are not members of the super-class. For example, a INSTRUCTOR_RESEARCHER can be a FACULTY, or a Grad_STUDENT, but not both. And all FACULTYs and Grad_STUDENTs are not INSTRUCTOR_RESEARCHERs.

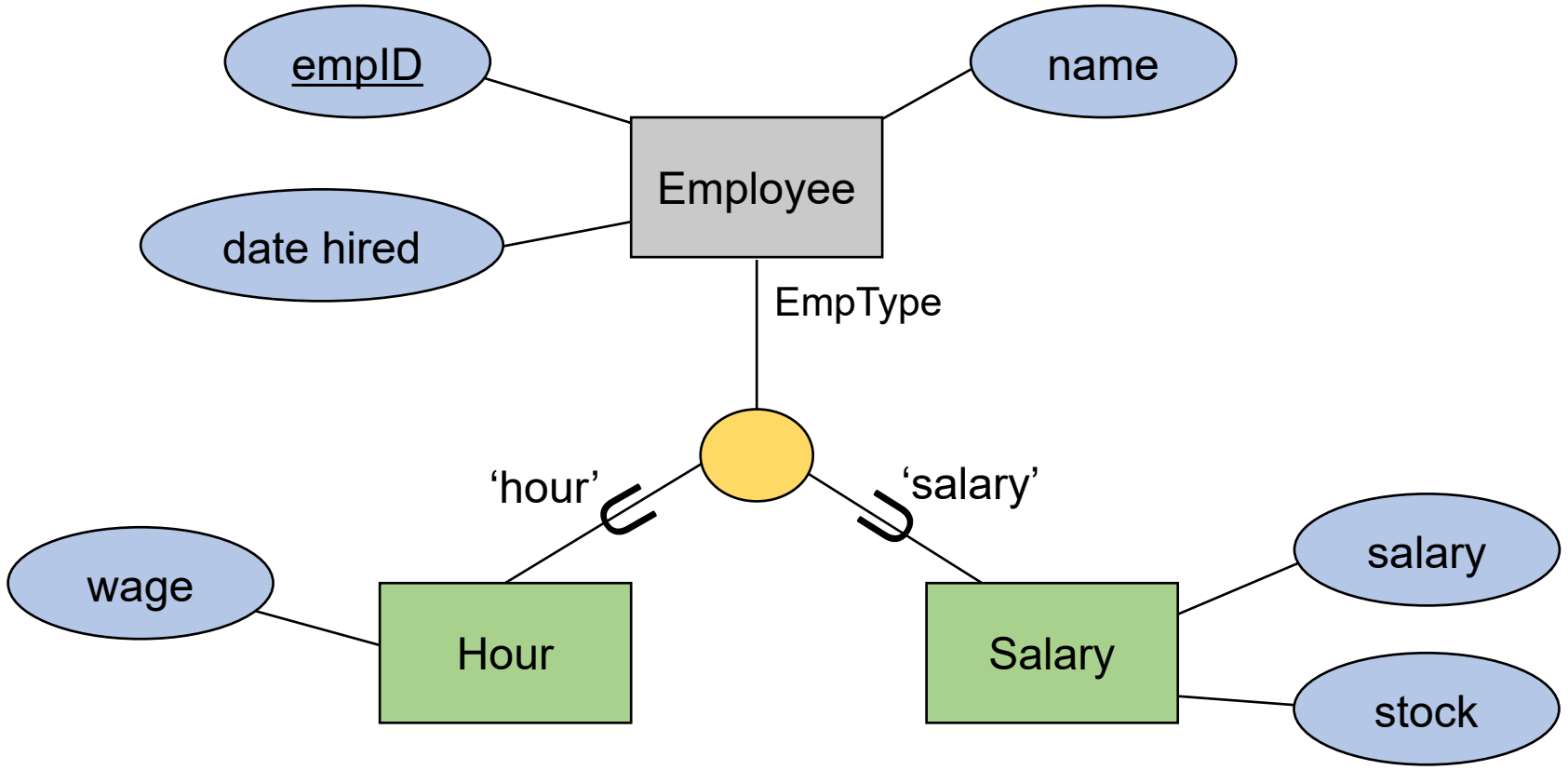


Multilevel Inheritance



4. Specify subtype discriminator and discriminator values

- **Subtype discriminator** – an attribute used to classify supertype's members into subtypes.



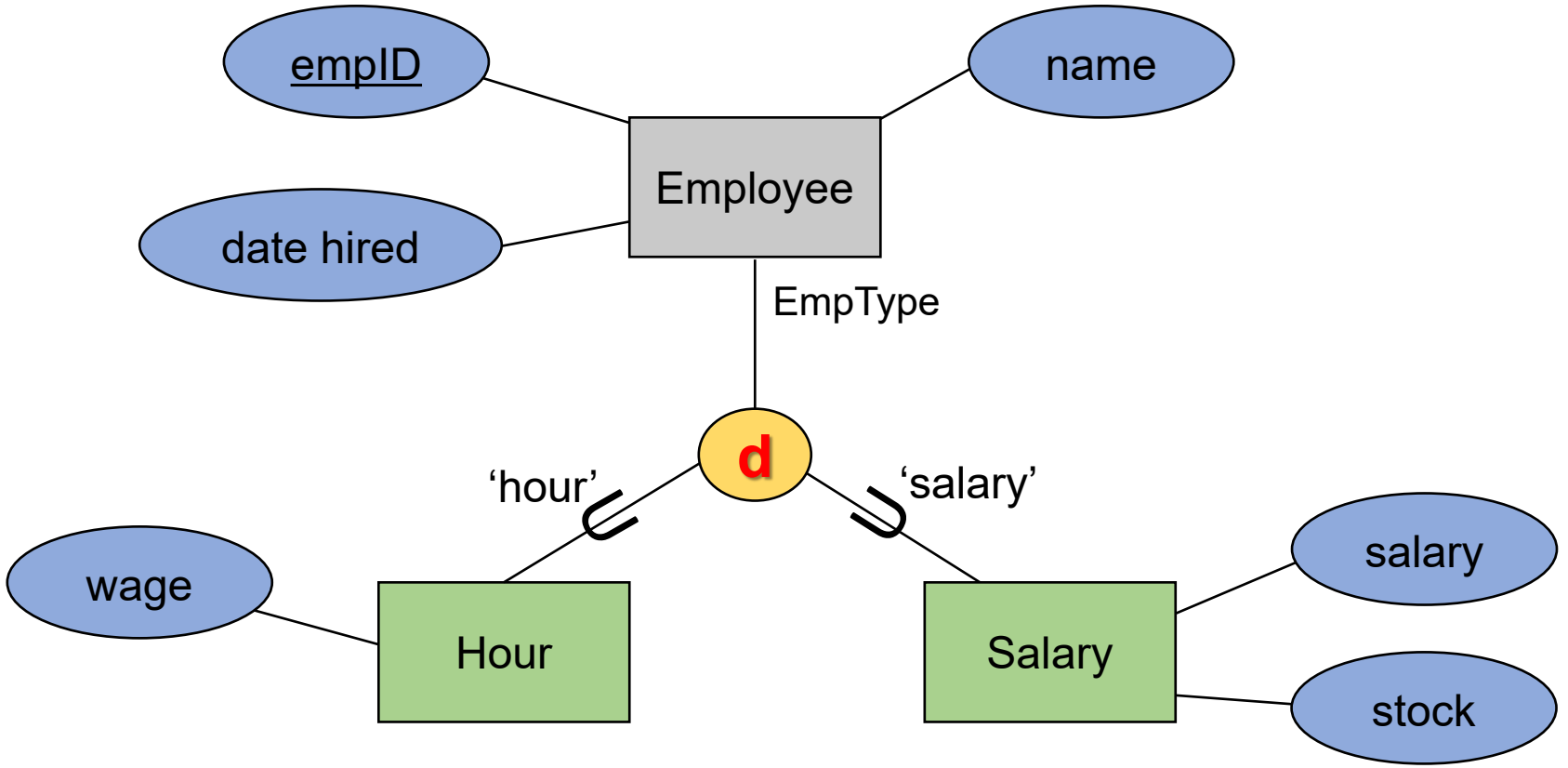
5. Identify discriminator constraints

- Two types of **Constraints:**
  - Disjointness
  - Completeness

5.1 Disjointness Constraints

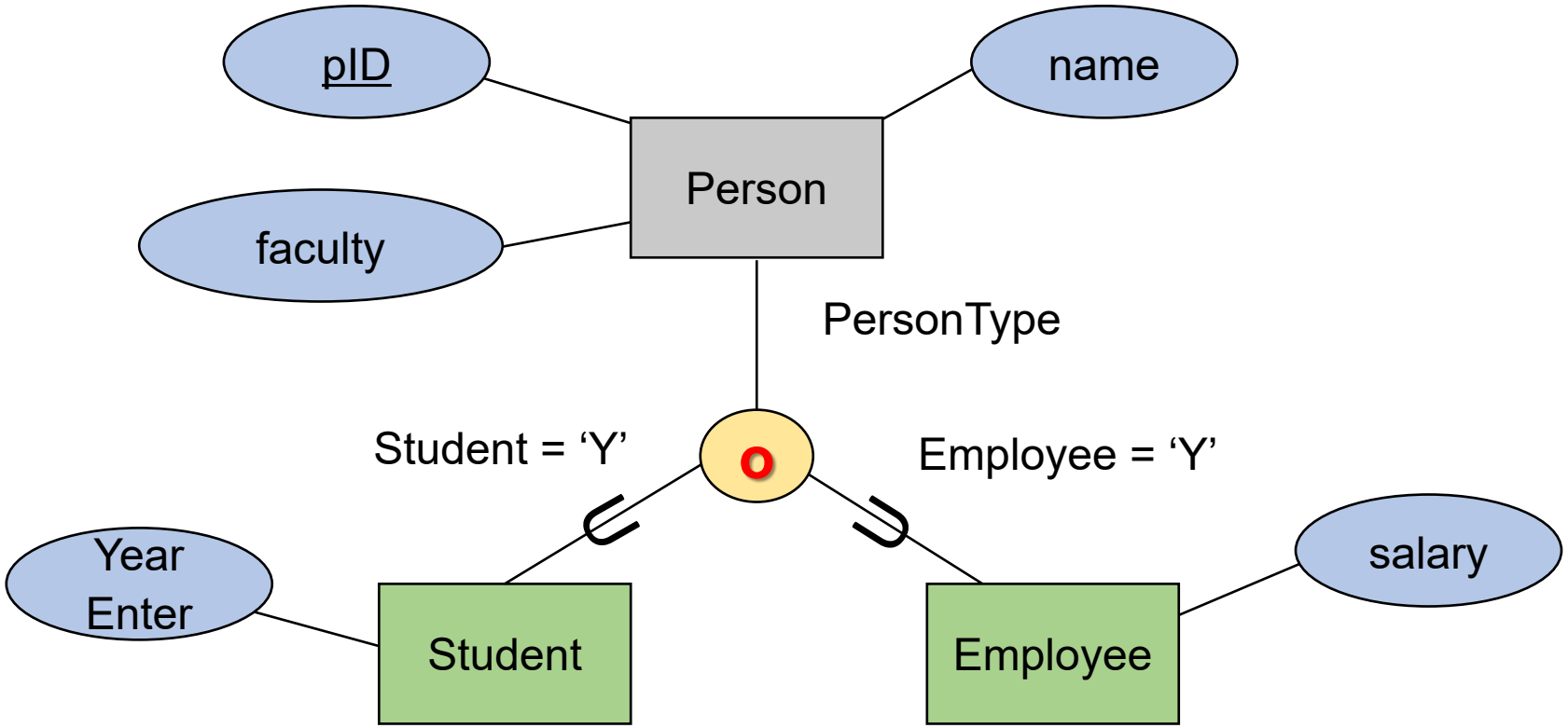
- **Disjoint Constraints** - the situation where an occurrence of a Supertype may also have more than one Subtype occurrence.
  - **Disjoint subtypes**
    - Also known as non-overlapping subtypes.
    - Subtypes that contain unique subset of supertype entity set.
  - **Overlapping subtypes**
    - Subtypes that contain non-unique subsets of supertype entity set.

5.1 Disjointness Constraints (cont.)



Disjoint subtype

5.1 Disjointness Constraints – (2)



Overlapping subtype

5.2 Completeness Constraints

- **Completeness Constraint** – deal with the issue of whether or not an occurrence of a Supertype must also have a corresponding Subtype occurrence.

5.2 Completeness Constraints – (2)

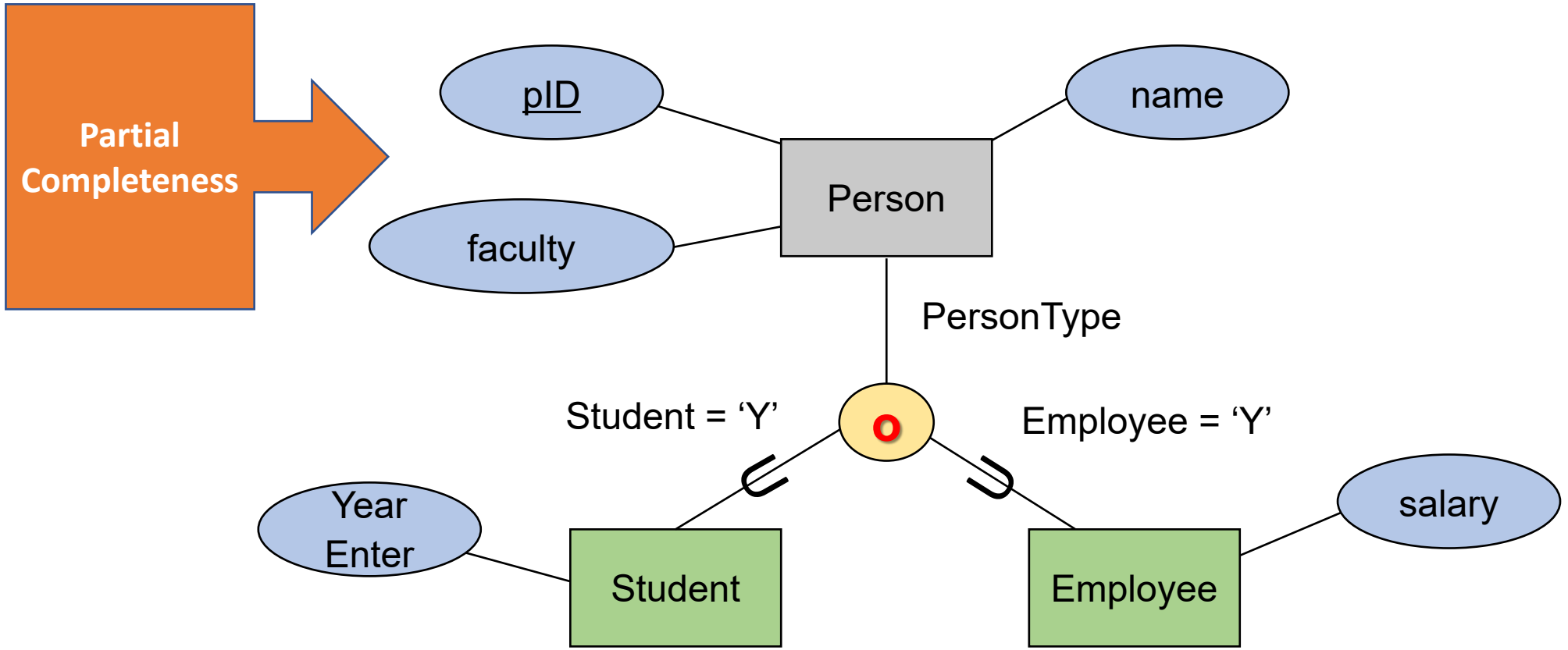
- **Total completeness**
  - Every supertype occurrence must be a member of at least 1 subclass.
  - Notation – Double line (as ER)
- **Partial completeness**
  - Some entities may belong to at least 1 subclass, or none at all.
  - Notation – Single line (as ER)

- **Example**
  - Students can be either graduate or undergraduate → total or partial?
  - Some employees may be students → total or partial?

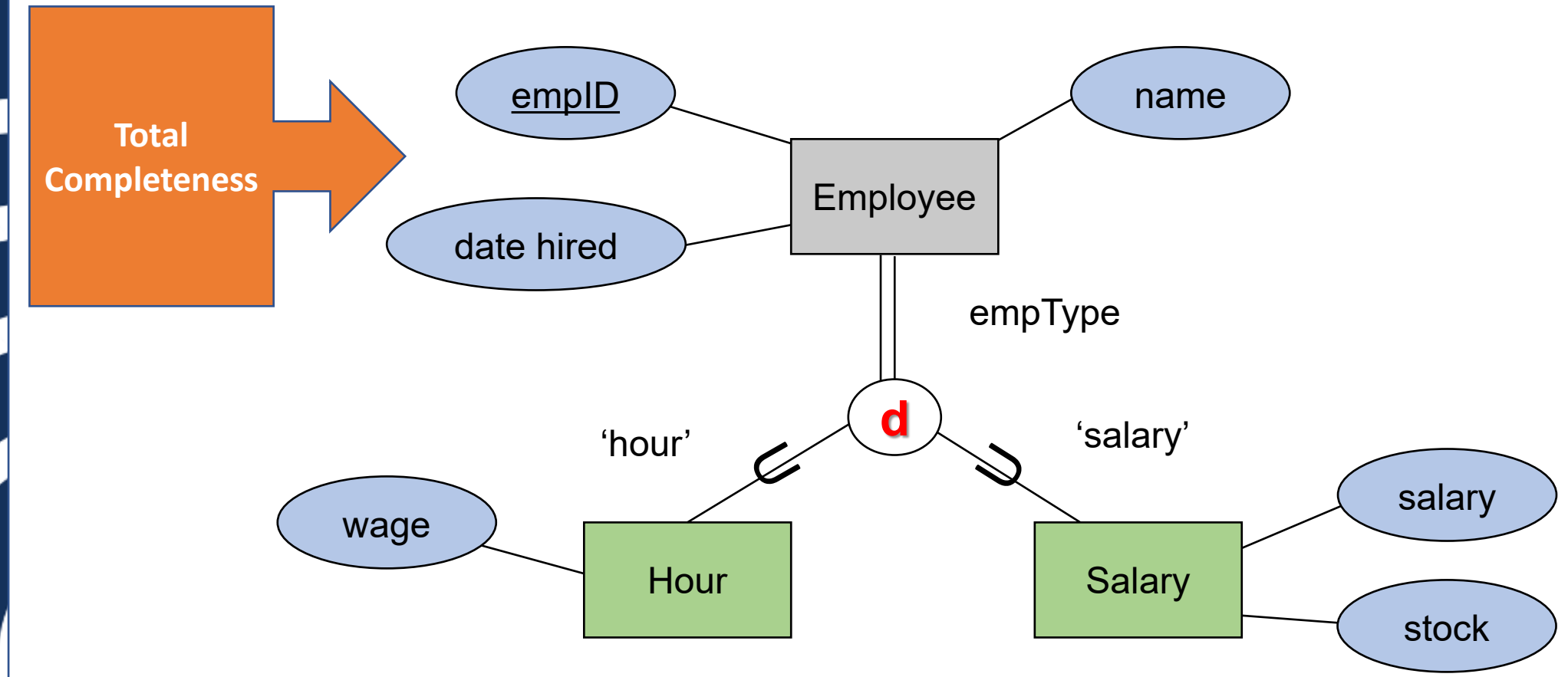
5.2 Completeness Constraints – (2)

- The generalization Person, of Man and Woman is **total** (the sets of men and the women constitute ‘all’ the people) **and exclusive** (a person is either a man or a woman).
- The generalization Vehicle of Automobile and Bicycle is **partial and exclusive**, because there are other types of vehicle (for example, motor bike) that are neither cars nor bicycle.
- The generalization Person of Student and Employee is **partial and overlapping**, because there are students who are also employed.

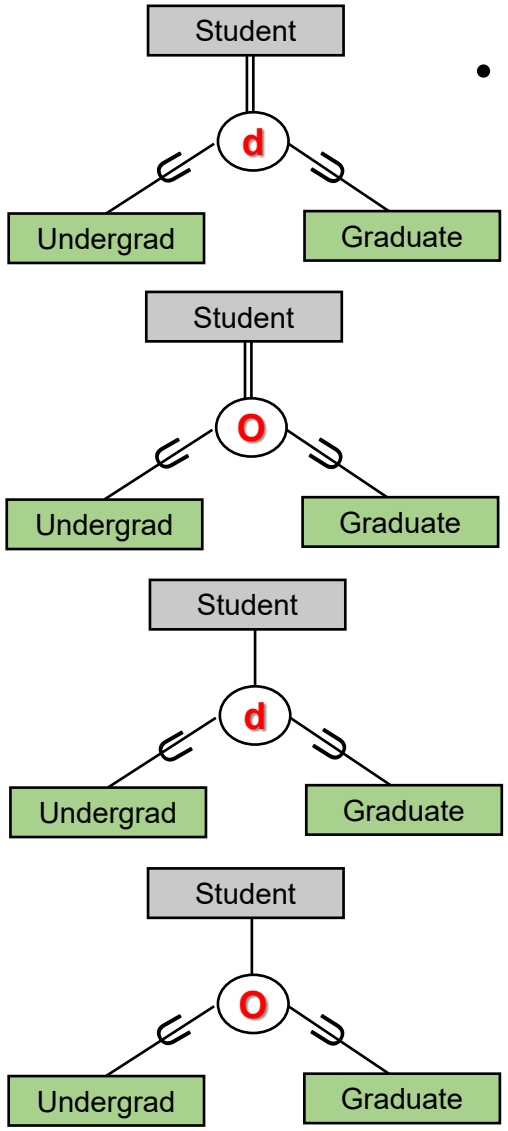
5.2 Completeness Constraints – (3)



5.2 Completeness Constraints – (4)



Disjointness VS Completeness



- Disjointness and Completeness are <u>independent</u>, so this yields 4 possibilities
  - **Total-Disjoint**
    - e.g., *All* students must be either undergrad or graduate students (each student can be only one type)
  - **Total-Overlapping**
    - e.g., *All* students must be undergrad, graduate or *both* types. (each student can be more than one type)
  - **Partial-Disjoint**
    - e.g., *Some* students can be either undergrad or graduate students. They *cannot* be both. However, some students can be other types such as prospective students.
  - **Partial-Overlapping**
    - e.g., *Some* students can be undergrad, graduate or *both* types. They can be more than one type. However, some students can be other types such as prospective students.

FIGURE 5.2 A SPECIALIZATION HIERARCHY

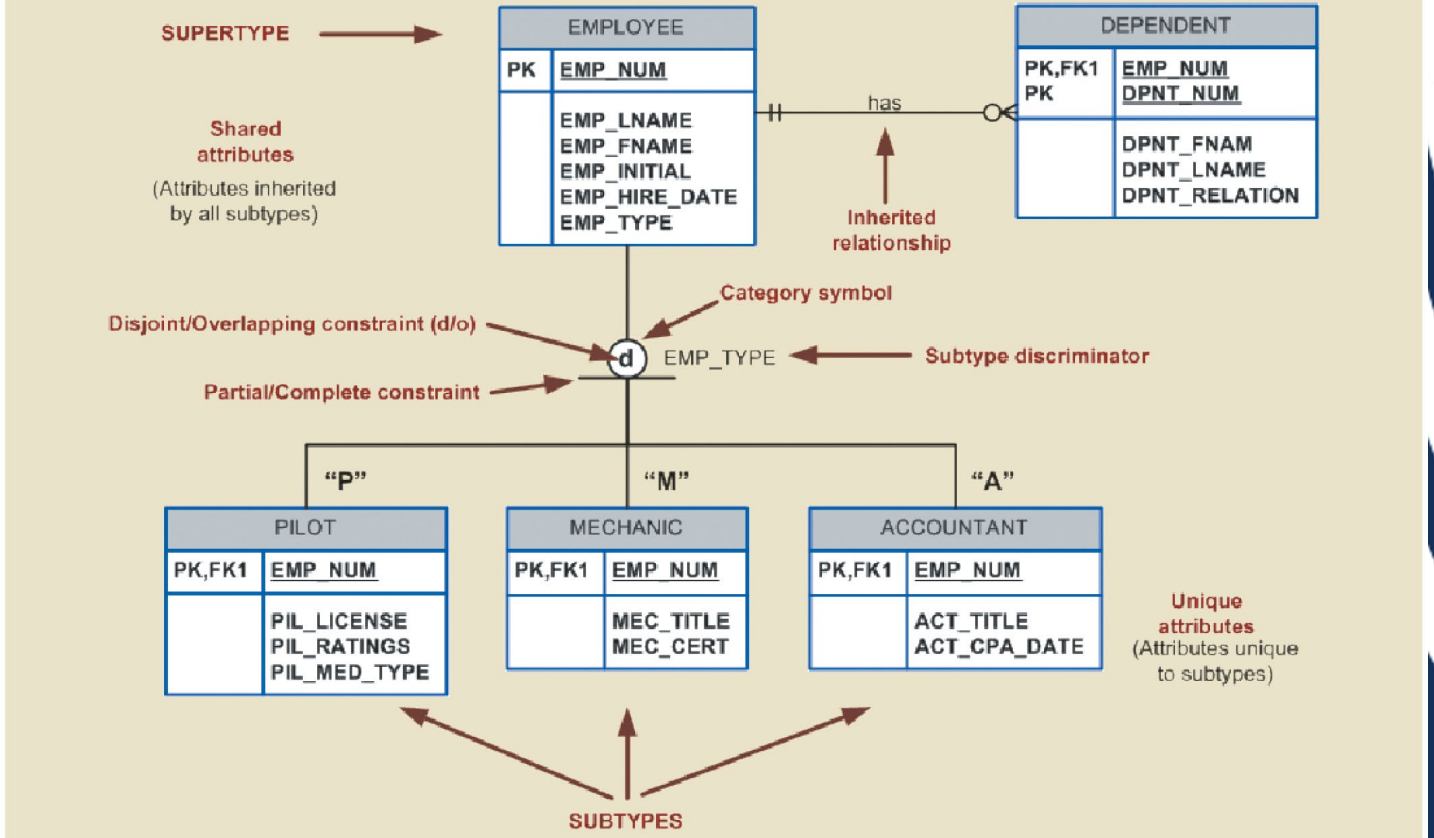
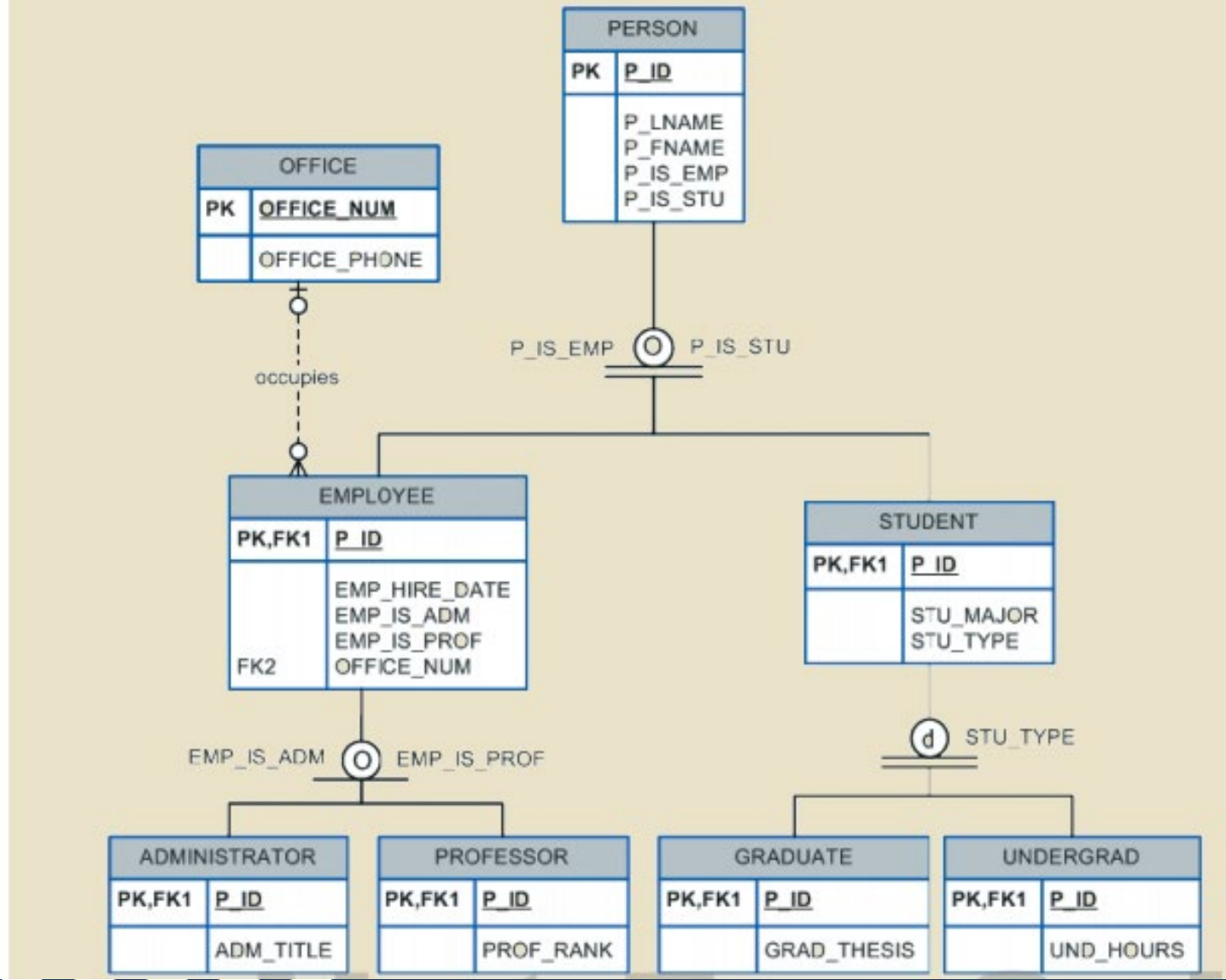


TABLE 5.2

SPECIALIZATION HIERARCHY CONSTRAINT SCENARIOS

TYPE	DISJOINT CONSTRAINT	OVERLAPPING CONSTRAINT
Partial 	Supertype has optional subtypes. Subtype discriminator can be null. Subtype sets are unique.	Supertype has optional subtypes. Subtype discriminators can be null. Subtype sets are not unique.
Total 	Every supertype occurrence is a member of only one subtype. Subtype discriminator cannot be null. Subtype sets are unique.	Every supertype occurrence is a member of at least one subtype. Subtype discriminators cannot be null. Subtype sets are not unique.

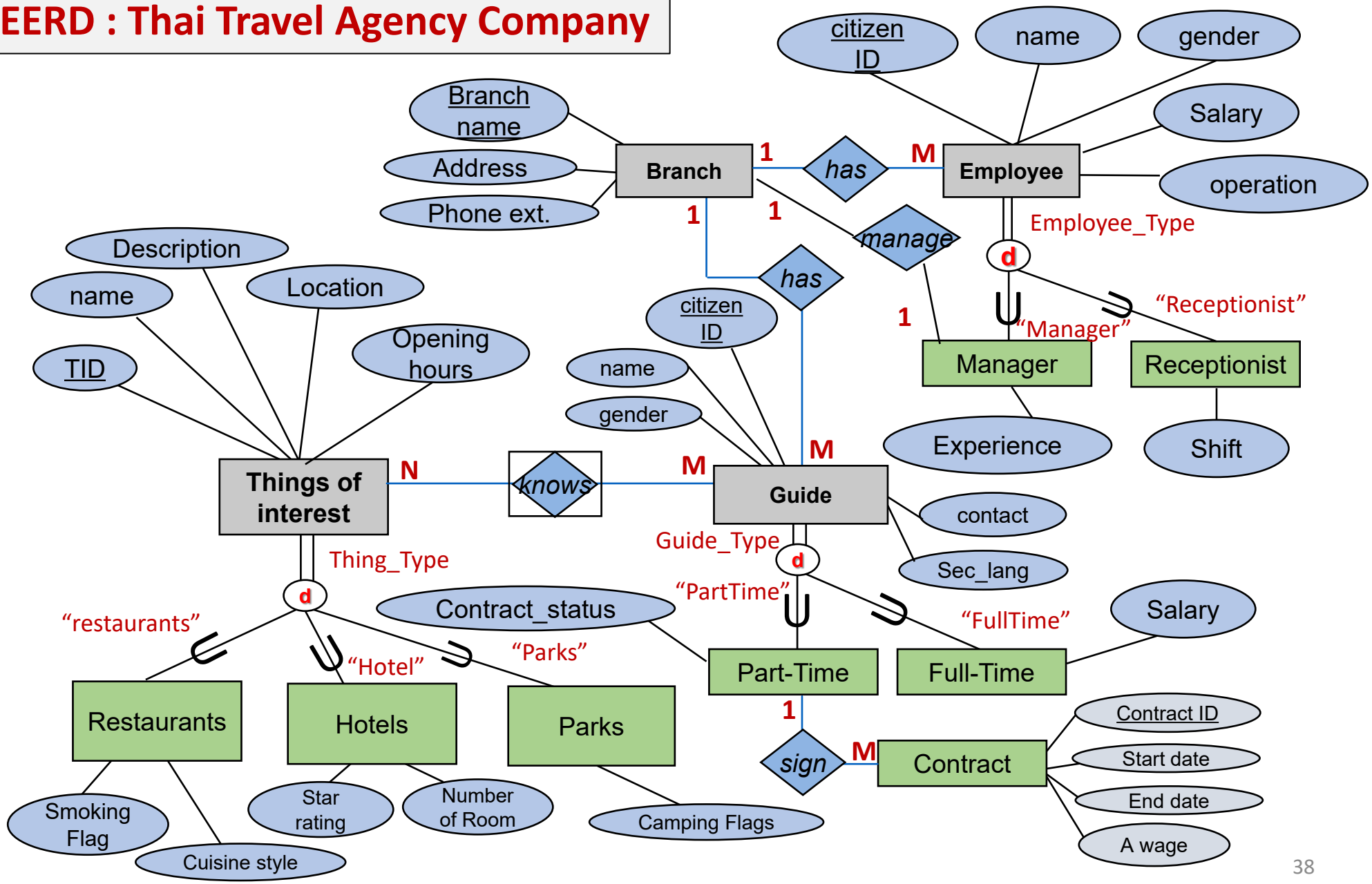
FIGURE 5.4 SPECIALIZATION HIERARCHY WITH OVERLAPPING SUBTYPES



Example:Thai Travel Agency Company

- The travel agency company has many branches. Each branch has a unique name, address, and telephone number extension. Each branch has a manager; and a manager will be hired to manage one and only one branch. A manager has a unique citizen ID, name, gender, salary and experience in year. Besides a manager, each branch has many receptionists to deal with the customers all the time. A receptionist also has a unique citizen ID, name, gender, salary, and shift (day, or night). Each branch also has general employees who work at the lowest level of operation (documenting, cleaning, and others). A general employee has a unique citizen ID, name, gender, salary
- Each branch also offers a guide for the customers. The company has two types of a guide which are a part time guide and a full time guide. Both types have a unique citizen ID, name, gender, contact, and second fluent language, but the full time guide has a salary. The part time guide has no specific salary, but the part time guide signs on a contract each time he/she is hired by the company. The contract has a unique contract number, start and end date, and a wage in hour-rating.
- A guide knows many things of interest. The thing of interest has a unique ID, name, description, location, and opening hours (opening and closing time). The thing of interest can be classified further into restaurants, hotels, and parks. The restaurant has smoking flag indicating that smoking is allowed or not, and cuisine style (Thai, Chinese, and etc.). The hotel has star rating, and number of rooms. The park has camping flag indicating that camping is allowed or not.

EERD : Thai Travel Agency Company



EERD : Thai Travel Agency Company

